

Outline

- 1) Convex Loss
- 2) Rao - Blackwell Theorem
- 3) UMVU Estimators
- 4) Examples

Unbiased Estimation

Recall strategies to choose an estimator

- 1) Summarize risk by a scalar (avg or sup)
- 2) Restrict to a smaller class of estimators

Today: Unbiased estimation estimator

Require $\mathbb{E}_{\theta} \delta(x) = g(\theta)$, $\forall \theta \in \Theta$

If we have complete sufficient stat $T(x)$,

- there is at most one unbiased $\delta^*(T(x))$

(If $\mathbb{E}_{\theta} \delta_1(T) = \mathbb{E}_{\theta} \delta_2(T) = g(\theta) \quad \forall \theta$ then $\delta_1 \equiv \delta_2$)

- if it exists, it uniformly minimizes risk for any convex loss function

Convex Loss Functions

Recall $f(x)$ is convex if, for all x_1, x_2 , all $\gamma \in [0, 1]$

$$f(\gamma x_1 + (1-\gamma)x_2) \leq \gamma f(x_1) + (1-\gamma)f(x_2)$$

strictly convex if $<$

Thm (Jensen) If f convex then
 $f(\mathbb{E}X) \leq \mathbb{E}f(X)$ for any r.v. X
if strictly convex then $<$ unless $X \stackrel{a.s.}{=} c$

Convex Loss $L(\theta, d)$ means convex in d

Ex. $L(\theta, d) = (g(\theta) - d)^2$

$$\begin{aligned} \text{MSE}(\theta; \delta) &= \mathbb{E}_{\theta} [(g(\theta) - \delta(x))^2] \\ &= \text{Bias}_{\theta}^2(\delta) + \text{Var}_{\theta}(\delta) \\ &= \text{Var}_{\theta}(\tau) \text{ if } \delta \text{ unbiased} \end{aligned}$$

Convex losses penalize us for making the estimator too noisy

Rao-Blackwell Theorem

Recipe to improve any $\delta(x)$ that violates the suff. principle.

Theorem (Rao-Blackwell)

Assume $T(X)$ sufficient, $\delta(X)$ estimator

$$\text{Let } \bar{\delta}(T(X)) = \mathbb{E} \left[\delta(X) \mid T(X) \right]$$

↖ no θ

If $L(\theta, \delta)$ convex then $R(\theta; \bar{\delta}) \leq R(\theta; \delta)$

If strictly convex then $R(\theta; \bar{\delta}) < R(\theta; \delta)$
unless $\delta(X) \stackrel{a.s.}{=} \bar{\delta}(T(X))$ for all θ

Proof

$$\begin{aligned} R(\theta; \bar{\delta}) &= \mathbb{E}_{\theta} \left[L(\theta, \mathbb{E}[\delta \mid T]) \right] \\ &\leq \mathbb{E}_{\theta} \mathbb{E} [L(\theta; \delta) \mid T] \\ &= R(\theta; \delta) \end{aligned}$$

< if strictly, unless $\delta \stackrel{a.s.}{=} \bar{\delta}$ $\square$

$\bar{\delta}(T)$ called the Rao-Blackwellization of $\delta(X)$

UMVU Estimators

Not all estimands have unbiased estimators:

Def We say $g(\theta)$ is U-estimable if $\exists \delta(x)$ with $E_{\theta} \delta = g(\theta) \forall \theta$

Def $\delta(x)$ is uniform minimum variance unbiased (UMVU) if for any unbiased $\tilde{\delta}$,
$$\text{Var}_{\theta}(\delta(x)) \leq \text{Var}_{\theta}(\tilde{\delta}(x)) \quad \forall \theta \in \Theta$$

Theorem For model $\mathcal{P} = \{P_{\theta} : \theta \in \Theta\}$, assume:

- i) $T(x)$ complete suff
- ii) $g(\theta)$ U-estimable

Then there exists a unique estimator of the form $\delta^*(T(x))$, which

- 1) is UMVU and minimizes
- 2) uniformly minimizes risk among all unbiased estimators

Proof

"All Rao-Blackwellizations lead to δ^* "

Existence

Take any δ_0 unbiased for $g(\theta)$

$$\text{Let } \delta^*(T) = \mathbb{E}[\delta_0 | T]$$

$$\mathbb{E}_\theta \delta^* = \mathbb{E}_\theta [\mathbb{E}[\delta_0 | T]] = \mathbb{E}_\theta \delta_0 = g(\theta)$$

Uniqueness

If $\delta(T)$ unbiased then

$$\mathbb{E}_\theta [\delta^*(T) - \delta(T)] = 0 \quad \forall \theta \in \Theta$$

$$\Rightarrow \delta^*(T) \stackrel{\text{a.s.}}{=} \delta(T) \quad (\text{Completeness})$$

Optimality wrt any convex loss

Suppose $\delta(X)$ unbiased, $\checkmark$ uniqueness

$$\text{let } \bar{\delta}(T) = \mathbb{E}[X | T] \stackrel{\text{a.s.}}{=} \delta^*(T)$$

Rao-Blackwell:

$$R(\theta; \delta^*) = R(\theta; \bar{\delta}) \leq R(\theta; \delta)$$

$$\text{Hence, } \text{MSE}(\theta; \delta^*) \leq \text{MSE}(\theta; \delta)$$

$$\text{" } \text{Var}_\theta(\delta^*) \leq \text{Var}_\theta(\delta)$$

So, δ^* UMVU



Finding the UMVUE

2 methods for finding UMVUE:

1) Find any unbiased estimator based on T

2) Find any unbiased estimator at all,
then R-B'ize it.

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Pois}(\theta)$, $g(\theta) = \theta^2$

$$p_{\theta}^{(1)}(x) = \frac{\theta^x e^{-\theta}}{x!} \quad \theta > 0, \quad x = 0, 1, \dots$$

Complete suff. stat $T(X) = \sum X_i \sim \text{Pois}(n\theta)$

$$p_{\theta}^T(t) = \frac{(n\theta)^t e^{-n\theta}}{t!}$$

$\delta(T)$ unbiased

$$\Leftrightarrow \sum_{t=0}^{\infty} \delta(t) p_{\theta}^T(t) = \theta^2, \quad \forall \theta$$

$$\Leftrightarrow \sum_{t=0}^{\infty} \delta(t) \frac{n^t}{t!} \theta^t = e^{n\theta} \theta^2 = \sum_{k=0}^{\infty} \frac{n^k}{k!} \theta^{k+2}, \quad \forall \theta$$

Match terms in power series:

$$\delta(0) = \delta(1) = 0, \quad \delta(t) = \frac{n^{t-2}}{(t-2)!} \cdot \frac{t!}{n^t} \quad t \geq 2$$

$$\Rightarrow \delta(T) = \frac{T(T-1)}{n^2} \quad (\approx \left(\frac{T}{n}\right)^2 \text{ for large } T)$$

Alternatively, we could R-Size $J_0(X) = X_1 X_2$

$$\mathbb{E}_\theta X_1 X_2 = (\mathbb{E}_\theta X_1)(\mathbb{E}_\theta X_2) = \theta^2$$

What is $\delta^*(T) = \mathbb{E}[X_1 X_2 | T]$?

$$X | T=t \sim \text{Multinom}(n, \frac{1}{n} \mathbf{1}_n)$$

$$X_1 | T=t \sim \text{Binom}(n, \frac{1}{n})$$

$$\Rightarrow \mathbb{E}[X_1 | T] = T/n$$

$$\text{Var}(X_1 | T) = T \cdot \frac{1}{n} (1 - \frac{1}{n}) = \frac{T(n-1)}{n^2}$$

$$\mathbb{E}[X_2 | T, X_1] = \frac{T - X_1}{n-1}$$

$$\begin{aligned} \mathbb{E}[X_1 X_2 | T] &= \mathbb{E}\left[X_1 \mathbb{E}[X_2 | T, X_1] | T\right] \\ &= \mathbb{E}\left[\frac{T}{n-1} X_1 - \frac{1}{n-1} X_1^2 | T\right] \\ &= \frac{T^2}{n(n-1)} - \frac{1}{n-1} \left(\frac{T^2}{n^2} + \frac{T(n-1)}{n^2} \right) \\ &= \frac{1}{n^2(n-1)} \left(T^2 n - T^2 - T(n-1) \right) \\ &= \frac{T(T-1)}{n^2} \end{aligned}$$

Ex $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U[0, \theta] \quad \theta > 0$

$T = X_{(n)}$ complete suff.

$$p_\theta^T = \frac{n}{\theta^n} t^{n-1} \mathbb{1}\{t \leq \theta\}$$

$$\mathbb{E}_\theta T = \int_0^\theta t \frac{n}{\theta^n} t^{n-1} dt = \frac{n}{n+1} \theta$$

$\Rightarrow \frac{n+1}{n} T$ is UMVU

Alternate $2X_1$ is unbiased

$$X_1 | T \sim \begin{cases} T & \text{wp } \frac{1}{n} \\ U[0, T] & \text{wp } \frac{n-1}{n} \end{cases}$$

$$\begin{aligned} \Rightarrow \mathbb{E}[2X_1 | T] &= 2T \cdot \frac{1}{n} + T \cdot \frac{n-1}{n} \\ &= \frac{n+1}{n} T \end{aligned}$$

Actually, $\frac{n+1}{n} T$ is inadmissible too!

Keener shows $\frac{n+2}{n+1} T$ has best MSE for any estimator $c \cdot T$.

Raises question: why do we require 0 bias?

Doubts about unbiasedness

UMVUE can be inadmissible, or just plain bad

Ex: $X \sim N_d(\mu, I_d)$ $\mu \in \mathbb{R}^d$

Estimate $g(\mu) = \|\mu\|^2$

$$\mathbb{E}_\mu \|X\|^2 = \sum_{j=1}^d \mathbb{E}(\mu_j + Z_j)^2 \quad Z_j \stackrel{\text{iid}}{\sim} N(0, 1)$$

$$= \sum \mu_j^2 + 1 = \|\mu\|^2 + d$$

$$\leadsto \delta(X) = \|X\|^2 - d \quad \text{is unbiased}$$

$$\Rightarrow \text{umvu (why?)}$$

But $\delta(X)$ can be negative...

$$\delta^*(X) = \max(0, \delta(X)) \quad \text{strictly dominates.}$$

Ex. $X \sim \text{Bin}(1000, \theta)$

Estimate $g(\theta) = P_\theta(X \geq 500)$

UMVUE is $1\{X \geq 500\}$ (why?)

Not reasonable...